

MEHANK

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Education

Vellore Institute of Technology

2022

B.Tech. in Computer Science, Minor in Cybersecurity and Digital Forensic

CGPA: 8.1

Work Experience

AI Engineering Intern

Oct 2025 – Nov 2025

Shell plc (Public Limited Company)

- Engineered a high-volume data processing pipeline in **Python**, utilizing **Unit Testing (PyTest)** and **CI/CD** workflows to ensure **99.9%** process continuity during intensive **ML** workloads.
- Standardized deployment workflows by authoring comprehensive technical documentation and **SOPs**, accelerating knowledge transfer and reducing future developer onboarding time.

Frontend Developer Intern

May 2025 – July 2025

Sahoo Engineering & Contractors

- Delivered a secure, production-ready application** by implementing **React**, **TypeScript strict mode** and **Zod validation**, enforcing input integrity.
- Optimized system performance to 60fps** via hardware acceleration and mitigating security vectors by systematically removing dead code and unused assets, ensuring resource efficiency and **codebase maintainability**.

Projects

RAG Document Q&A System — *Flask, ChromaDB, Ollama, NLP, LangChain*

[View project](#)

Jan 2026

- Built Architected a full-stack **Agentic RAG solution** using **LangChain** and **Flask**, **ChromaDB**, and **Ollama** to process multi-format documents, implementing semantic chunking and embedding-based cosine similarity to optimize context retrieval for **LLM Agents**, while deploying anti-hallucination safeguards with **20%+** vocabulary overlap validation to ensure grounded, accurate Q&A.

Legacy Booking System Security — *Laravel, PHP, MySQL, Next.js, RESTful API*

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Dec 2025

- Modernized a legacy PHP booking system by migrating to **Laravel** and **PHP**, remediating **15+** SQL injection vulnerabilities through Eloquent ORM and prepared statements while maintaining **100%** backward compatibility for a **Next.js** and **TypeScript** frontend; executed a **zero-downtime** migration that significantly enhanced the security posture and reliability of the **MySQL** database and RESTful API layer.

Rooftop Images Segmentation — *Python, PyTorch, CI/CD, TensorFlow*

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Nov 2025

- Built and deployed a production-ready **deep learning** image segmentation pipeline using **Python**, **PyTorch**, and **TensorFlow**, processing **1,600+** rooftop images with automated validation, checkpointing, and **CI/CD**-aligned evaluation workflows, reducing manual intervention by **80%** and improving model performance by **2x**.

SuPOR Cipher Suite — *Python, NumPy, Modular Architecture*

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May 2025

- Programmed a modular **stream-cipher** suite using **Python** and **NumPy**, integrating AES-inspired S-box substitution and **64-bit** seeded permutations to achieve **100%** byte-level data integrity, while deploying a high-fidelity CustomTkinter **glassmorphic** UI for seamless, secure cross-platform file handling.

Real-time Sign Language Recognition — *Kotlin, TensorFlow Lite, YOLOv8, CameraX*

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Feb 2025

- Developed a real-time **Android** application using **Kotlin** and MVVM to recognize **26+ ASL gestures**, achieving **35 FPS** and **80ms** latency via **YOLOv8** edge inference while optimizing **CameraX** pipelines and NMS thresholds to increase detection accuracy by **40%** across high-variability inputs.

Packet Sniffer Network Analysis Tool — *Python, Flask, Scapy*

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Mar 2024

- Engineered a multi-threaded network analyzer using **Python** and **Scapy** for **real-time** Deep Packet Inspection of HTTP/TLS traffic.

Technical Skills

Programming Languages: Python, Go, Java, PHP, SQL, HTML, CSS, JavaScript

Frameworks: PyTorch, Tensorflow, Docker, Git, AWS, PostgreSQL, Node.js, FastAPI, React.js, LangChain, PyTest

Concepts: REST APIs, Wireshark, Nmap, IDS/IPS, IAM, Burp Suite, OWASP, Cipher, Convolutional Neural Networks, Large Language Models

Achievements

- Rank<1000 in TCS CodeVita**(Round 2 Qualifier); Participant in **Samsung Solve for Tomorrow** and **NHAI Hackathons**.
- Earned Certifications in **Cybersecurity & DevOps (IBM)**, **Google Cybersecurity**, and **Software Engineering (HackerRank)**.
- Achieved **5-Star Gold Badge in SQL** (HackerRank) and **Linux Badge** (TryHackMe); Scored **100/100 in Mathematics** (CBSE Boards).